

Nikunj Pradhan

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EDUCATION

Missouri University of Science and Technology

Masters of Science: Computer Science

Bachelor of Science: Computer Science

GPA: 4.0/4.0

Aug 2025 - May 2027

Aug 2023 - Dec 2026

EXPERIENCE

Data Science Intern

Quest Analytics

May 2026 – Present

Overland Park, KS

- Standardized 12M+ Medicaid provider records from 20+ state sources through a ETL Medallion pipeline, producing a unified gold dataset containing 3M providers.
- Resolved 3,000+ provider records by building a LangGraph-based research tool for entity resolution, automating iterative search, extraction, and validation while routing tasks across LLMs to optimize inference cost and latency.
- Expanded internal Medicaid provider coverage by 42% by identifying 500,000+ previously absent providers and creating a high-confidence dataset for analytics and data science teams.
- Architected a data preprocessing pipeline utilizing fuzzy matching and semantic embeddings to unify 20+ heterogeneous data sources, with UUID-based lineage tracking.

Data Science Co-Op

Hunter Engineering Company

Jan 2025 – Aug 2025

Bridgeton, MO

- Boosted end-to-end classification accuracy by 16% (68% to 84%) by optimizing YOLO-based pipelines to maintain high performance across degraded image quality and highly imbalanced regional datasets.
- Processed 2M+ images from 600+ worksites by building and maintaining ETL pipelines that ingested image data and metadata into SQL Server and AWS S3.
- Reduced geometric correction error by 23% by developing a ResNet-based regression model that estimated four corner coordinates for perspective rectification of distorted regions.
- Delivered ML infrastructure tools including model performance dashboards, dataset analytics platforms, and AI-assisted annotation services to support R&D Department.

Research Assistant

Missouri University of Science and Technology

Aug 2023 – Dec 2024

Rolla, MO

- Improved point-tracking accuracy by 62% by replacing deterministic OpenCV methods with a custom deep learning model to reliably track soil features and quantify deformation during compression.
- Developed a 3D reconstruction pipeline using C++, implementing watershed segmentation and ray-tracing triangulation to enable the generation of precise soil compression models for civil engineering research.

PROJECTS

Nepali Devanagari OCR Pipeline | *PyTorch, Python, FastAPI, Docker, ONNX*

- Optimized CPU OCR inference with ONNX Runtime, processing batches of 8 document images in 3s via a Dockerized FastAPI service; developed a custom CRNN-based OCR pipeline achieving 7.1% CER and 16.5% WER.

FIFA World Cup Prediction & Simulation | *XGBoost, Scikit-learn, Pandas, NumPy*

- Boosted match prediction accuracy from 58% to 64% with an XGBoost forecasting pipeline; applied temperature scaling for calibrated probabilities (0.19 mean RPS), and leveraged Monte Carlo simulations to forecast tournament outcomes.

Movie Recommendation & Watchlist Platform | *FastAPI, React, PostgreSQL, Scikit-learn, Kubernetes*

- Deployed a full-stack movie watchlist platform with personalized recommendations using TF-IDF and cosine similarity for content-based filtering alongside ALS collaborative filtering, with per-user caching to reduce latency.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++

Machine Learning & AI: PyTorch, Scikit-learn, XGBoost, Hugging Face, LangChain, RAG, MLflow

Data Engineering: Apache Spark (PySpark), Spark SQL, Delta Lake, Apache Kafka, Databricks

Data Analysis & Visualization: Pandas, NumPy, OpenCV, Seaborn

Backend & APIs: FastAPI, Spring Boot, Redis

Databases & Storage: PostgreSQL, pgvector, SQL Server, DynamoDB, MinIO

Cloud & DevOps: AWS, Docker, Kubernetes, Git, CI/CD, GitHub Actions